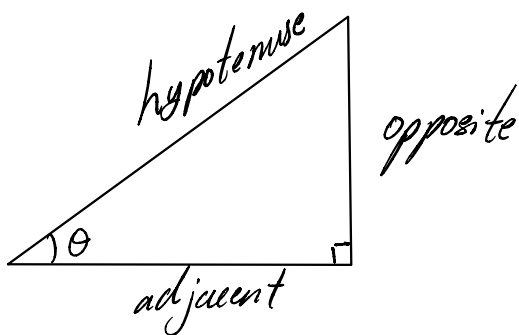


Lecture 8, Jan 8, 2025.

Appendix D (trigonometry).

The trigonometric functions.



$$\sin \theta = \frac{\text{opp}}{\text{hyp}}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}}$$

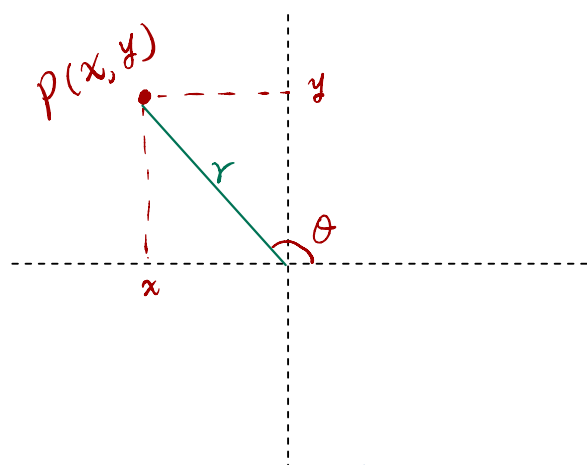
$$\tan \theta = \frac{\text{opp}}{\text{adj}}$$

$$\csc \theta = \frac{\text{hyp}}{\text{opp}}$$

$$\sec \theta = \frac{\text{hyp}}{\text{adj}}$$

$$\cot \theta = \frac{\text{adj}}{\text{opp}}$$

If $P = (x, y)$



$$\sin \theta = \frac{y}{r}$$

$$\cos \theta = \frac{x}{r}$$

$$\tan \theta = \frac{y}{x}$$

$$\csc \theta = \frac{r}{y}$$

$$\sec \theta = \frac{r}{x}$$

$$\cot \theta = \frac{x}{y}$$

trigonometric identities.

i). $\csc \theta = \frac{1}{\sin \theta}$; $\sec \theta = \frac{1}{\cos \theta}$; $\cot \theta = \frac{1}{\tan \theta}$; $\tan \theta = \frac{\sin \theta}{\cos \theta}$; $\cot \theta = \frac{\cos \theta}{\sin \theta}$.

ii). $\sin^2 \theta + \cos^2 \theta = 1$.

iii). $\tan^2 \theta + 1 = \sec^2 \theta$.

iv). $1 + \cot^2 \theta = \csc^2 \theta$.

v). $\sin(-\theta) = -\sin \theta$.

vi). $\cos(-\theta) = \cos(\theta)$

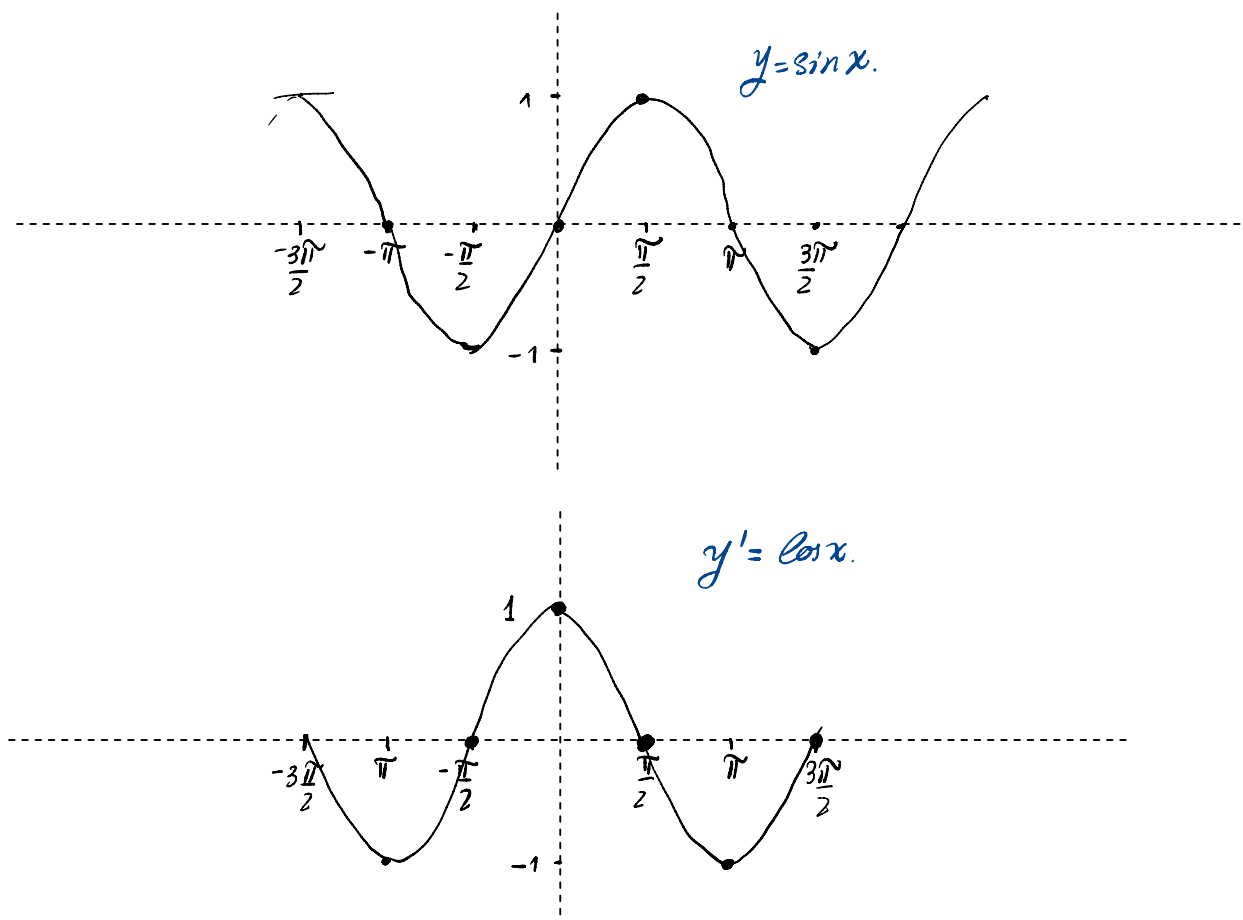
vii). • $\sin(x+y) = \sin x \cos y + \sin y \cos x$.

• $\sin(x-y) = \sin x \cos y - \sin y \cos x$.

- $\cos(x+y) = \cos x \cos y - \sin x \sin y$.
- $\cos(x-y) = \cos x \cos y + \sin x \sin y$.

viii). $\sin 2x = 2 \sin x \cos x$.

2.4. Derivatives of trigonometric functions.



Claim. if $f(x) = \sin x$ then $f'(x) = \cos x$.

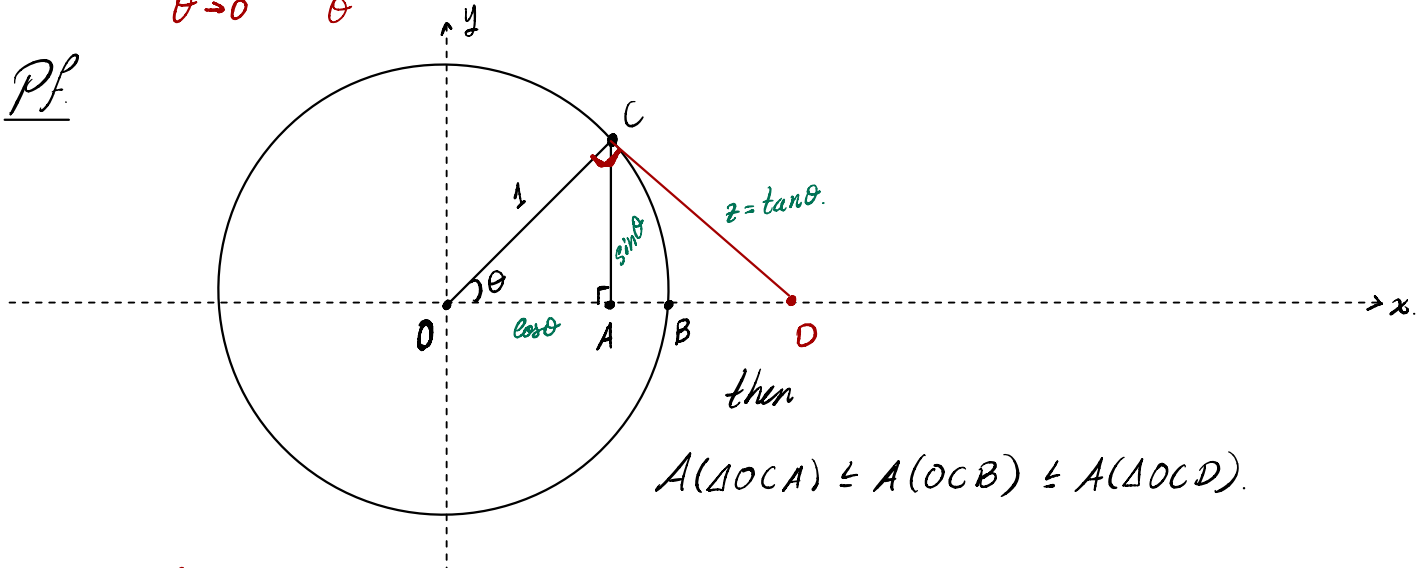
$$\begin{aligned}
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sin x \cos h + \sin h \cos x - \sin x}{h} \\
 &= \lim_{h \rightarrow 0} \left[\frac{\sin x \cos h - \sin x}{h} + \frac{\sin h \cos x}{h} \right] \\
 &= \lim_{h \rightarrow 0} \left[\sin x \frac{(\cos h - 1)}{h} + \cos x \cdot \frac{\sin h}{h} \right]
 \end{aligned}$$

$$= \lim_{h \rightarrow 0} \sin x \frac{\cos h - 1}{h} + \lim_{h \rightarrow 0} \cos x \frac{\sin h}{h}$$

$$= \sin x \cdot \underbrace{\lim_{h \rightarrow 0} \frac{\cos h - 1}{h}}_0 + \cos x \underbrace{\lim_{h \rightarrow 0} \frac{\sin h}{h}}_1$$

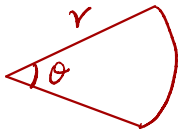
Thm. $\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1.$

Pf.



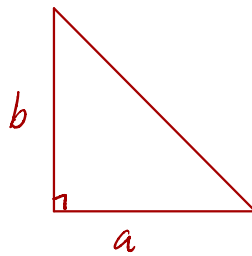
$$A(\triangle OCA) \leq A(\text{sector } OCB) \leq A(\triangle OCD).$$

Area of a circular sector.



$$A = \frac{\theta \cdot r^2}{2}$$

Area of a triangle.



$$A = \frac{a \times b}{2}$$

$$A(\triangle OCA) = \frac{\cos \theta \sin \theta}{2} ; A(\text{sector } OCB) = \frac{\theta \cdot 1^2}{2} = \frac{\theta}{2} ; \text{Area}(\triangle OCD) = \frac{\tan \theta}{2}$$

then $\frac{\cos \theta \sin \theta}{2} \leq \frac{\theta}{2} \leq \frac{\tan \theta}{2}$

$$\cos \theta \sin \theta \leq \theta \leq \frac{\sin \theta}{\cos \theta} \quad \text{by } a \leq b \Rightarrow \frac{1}{b} \leq \frac{1}{a}$$

$$\frac{\cos \theta}{\sin \theta} \leq \frac{1}{\theta} \leq \frac{1}{\cos \theta \sin \theta}$$

$$\cos \theta \leq \frac{\sin \theta}{\theta} \leq \frac{1}{\cos \theta}$$

and $\lim_{\theta \rightarrow 0} \cos \theta = 1$, $\lim_{\theta \rightarrow 0} \frac{1}{\cos \theta} = 1$. By S-T we get.

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1.$$

Thm. $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$

This implies that

$$f'(x) = \sin x \lim_{h \rightarrow 0} \frac{\cos h - 1}{h} + \cos x \lim_{h \rightarrow 0} \frac{\sin h}{h} = \cos x$$

Example. Differentiate $y = x^2 \sin x$.

Sln. $y = f(x) \cdot g(x)$ where $f(x) = x^2$, $g(x) = \sin x$
 $f'(x) = 2x$ $g'(x) = \cos x$.

Using the product rule

$$y' = 2x \sin x + x^2 \cos x.$$

Thm. $\frac{d}{dx} (\cos x) = -\sin x$.

Example. Compute $\frac{d}{dx} (\tan x)$.

Sln.

$$\begin{aligned} \frac{d}{dx} (\tan x) &= \frac{d}{dx} \left(\frac{\sin x}{\cos x} \right) = \frac{\frac{d}{dx} (\sin x) \cdot \cos x - \sin x \cdot \frac{d}{dx} (\cos x)}{\cos^2 x} \\ &= \frac{\cos x \cdot \cos x - \sin x (-\sin x)}{\cos^2 x} \\ &= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x. \end{aligned}$$

Table of derivatives.

$$\frac{d}{dx} (\sin x) = \cos x \quad ; \quad \frac{d}{dx} (\tan x) = \sec^2 x \quad ; \quad \frac{d}{dx} (\sec x) = \sec x \cdot \tan x$$

$$\frac{d}{dx} (\cos x) = -\sin x \quad ; \quad \frac{d}{dx} (\csc x) = -\csc x \cot x \quad ; \quad \frac{d}{dx} (\cot) = -\csc^2 x.$$

Example. if $f(x) = \frac{\sec x}{1 + \tan x}$ compute f' . For what values of x does the graph of f have horizontal tangent.

Sln. $f'(x) = \frac{\frac{d}{dx}(\sec x) \cdot (1 + \tan x) - \sec x \cdot \frac{d}{dx}(1 + \tan x)}{(1 + \tan x)^2}$

$$= \frac{\sec x \cdot \tan x (1 + \tan x) - \sec x (\sec^2 x)}{(1 + \tan x)^2}$$

$$= \frac{\sec x (\tan^2 x + \tan x - \sec^2 x)}{(1 + \tan x)^2}$$

$$\tan^2 x + 1 = \sec^2 x$$

$$\tan^2 x - \sec^2 x = -1$$

$$= \frac{\sec x (\tan x - 1)}{(1 + \tan x)^2}$$

Notice that $\sec x \neq 0$ then $f'(x) = 0$ when $\tan x = 1$ or,

$$x = \frac{\pi}{4} + n\pi \text{ where } n \in \mathbb{Z}.$$

$$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

and

$$\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0.$$

Example. find $\lim_{x \rightarrow 0} \frac{\sin 7x}{4x}$.

Sln. $\lim_{x \rightarrow 0} \frac{\sin 7x}{4x} = \lim_{x \rightarrow 0} \frac{7 \sin 7x}{4(7x)} = \frac{7}{4} \lim_{x \rightarrow 0} \frac{\sin 7x}{7x} = \frac{7}{4}$

Example. Calculate $\lim_{x \rightarrow 0} x \cot x$.

Sln. $\lim_{x \rightarrow 0} x \cot x = \lim_{x \rightarrow 0} x \frac{\cos x}{\sin x} = \lim_{x \rightarrow 0} \frac{\cos x}{\frac{\sin x}{x}} = \frac{\lim_{x \rightarrow 0} \cos x}{\lim_{x \rightarrow 0} \frac{\sin x}{x}} = \frac{1}{1} = 1$

Example. Calculate $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\sin \theta}$

Sln. $\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\sin \theta} = \lim_{\theta \rightarrow 0} \frac{\frac{\cos \theta - 1}{\theta}}{\frac{\sin \theta}{\theta}} = \frac{\lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta}}{\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta}} = \frac{0}{1} = 0$

2.5 The Chain Rule.

Suppose that you are asked to compute f' where

$$f(x) = \sqrt{x^2 + 1}$$

Notice that $g(x) = \sqrt{x}$ the outer function and $h(x) = x^2 + 1$ is the inner function.

and $f(x) = (g \circ h)(x)$.

The Chain Rule: If g is differentiable at x and f is differentiable at $g(x)$ then $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x and

$$F'(x) = f'(g(x)) \cdot g'(x).$$

Example. Notice that $F(x) = \sqrt{x^2 + 1}$ if $f(x) = \sqrt{x}$, $g(x) = x^2 + 1$ and $F(x) = f(g(x))$. then by the Chain Rule.

$$F'(x) = f'(g(x)) \cdot g'(x)$$

$$f'(x) = \frac{1}{2\sqrt{x}} ; g'(x) = 2x.$$

$$F'(x) = \frac{1}{2\sqrt{x^2 + 1}} \cdot (2x) = \frac{x}{\sqrt{x^2 + 1}}.$$

Rmk. $\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$

outer function evaluated at inner function derivative of outer function evaluated at inner function derivative of inner function.

Example. Compute y' if

a). $y = \sin(x^2)$

b). $y = \sin^2(x)$.

Sln. a). $y = f(g(x))$ where $f(x) = \sin x$, $g(x) = x^2$.

$$f'(x) = \cos x \quad g'(x) = 2x$$

$$y' = \cos(x^2) \cdot 2x = 2x \cos(x^2).$$

b). $y = f(g(x))$ where $f(x) = x^2$ $g(x) = \sin x$.
 $f'(x) = 2x$ $g'(x) = \cos x$

$$y' = 2 \sin x \cdot \cos x.$$

Thm. The power rule combined with the Chain Rule. if n is any real number and $u = g(x)$ then

$$\frac{d}{dx}(u^n) = \frac{d}{dx}(g^n(x)) = n g^{n-1}(x) \cdot g'(x).$$

Example. Compute y' where $y = (x^3 - 1)^{100}$

Sln. $y = f(g(x))$ $f = x^{100}$ $g(x) = x^3 - 1$ $f'(x) = 100 x^{99}$
 $g'(x) = 3x^2$

$$y' = 100(x^3 - 1)^{99} \cdot 3x^2.$$

Example. Find $f'(x)$ if $f(x) = \frac{1}{\sqrt[3]{x^2 + x + 1}}$

Sln. $f(x) = g(h(x))$ $g(x) = x^{-1/3}$; $h(x) = x^2 + x + 1$.

$$g'(x) = -\frac{1}{3} x^{-1/3 - 1} = -\frac{1}{3} x^{-4/3} \quad h'(x) = 2x + 1.$$

$$f'(x) = -\frac{1}{3} (x^2 + x + 1)^{-4/3} (2x + 1).$$

Example. • Find g' if $g(t) = \left(\frac{t-2}{2t+1}\right)^9$
 • Compute y' if $y = (2x+1)^5 (x^3 - x + 1)^4$ } for you!

Example. If $F(x) = \sin(\cos(\tan x))$ compute F' .

Sln. $F(x) = f(g(x))$ where $f(x) = \sin x$ $g(x) = \cos(\tan x)$
 $g(x) = h(s(x))$ where $h(x) = \cos x$ $s(x) = \tan x$.

By the chain Rule.

$$f'(x) = f'(g(x)) \cdot g'(x) \quad \text{but} \quad g'(x) = h'(g(x)) g'(x).$$

$$f'(x) = \cos x ; \quad h'(x) = -\sin x \quad g'(x) = \sec^2 x.$$

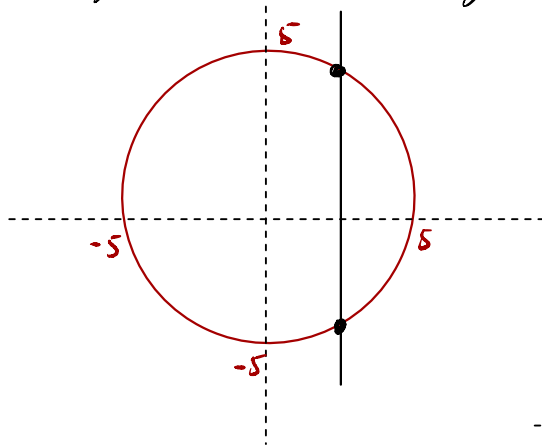
$$\begin{aligned} f'(x) &= \cos(\cos(\tan x)) \cdot (-\sin(\tan x)) \sec^2 x \\ &= -\cos(\cos(\tan x)) \sin(\tan x) \sec^2 x. \end{aligned}$$

Problem. Compute y' if $y = \sqrt{\sec(x^3)}$. *For you!*

We are study functions like

$$y = \sqrt{x^3 + 1} \quad y = x \sin x.$$

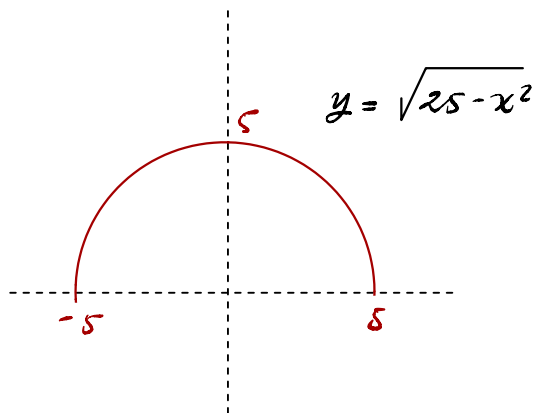
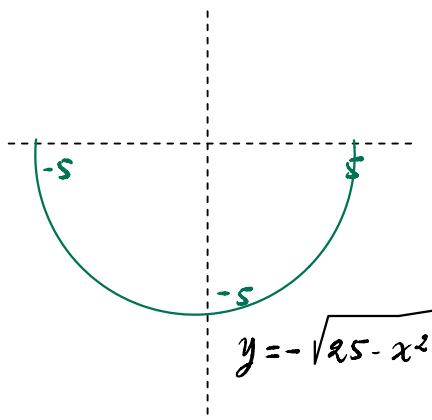
Now, consider $x^2 + y^2 = 25$. then.



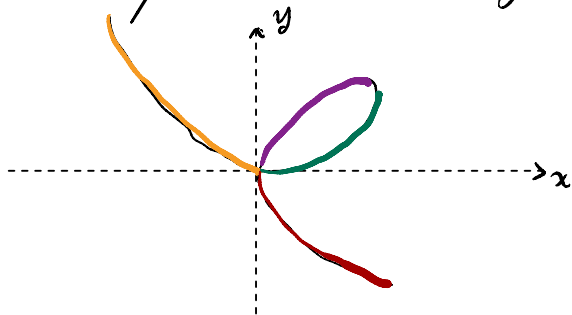
$$x^2 + y^2 = 25$$

$$y^2 = 25 - x^2$$

$$y = \pm \sqrt{25 - x^2}$$



for example let $x^3 + y^3 = 6xy$.



Implicit Differentiation. We don't need to solve an equation for y in terms of x in order to find the derivative of y .

Instead we can use implicit differentiation. This consists of differentiating both sides of the equation with respect to x and then solving the resulting equation for $\frac{dy}{dx}$.

Example. if $x^2 + y^2 = 25$ find $\frac{dy}{dx}$ and find the equation of the tangent to the circle at $(3, 4)$.

Sol. Step 1. Take $\frac{d}{dx}$ in both sides.

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25)$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0.$$

Step 2. if you need compute $\frac{d}{dx}(G(y))$ use the chain rule.

$$2x + \frac{d}{dx}(y^2) = 0$$

$$\text{Let } G(y) = y^2$$

$$G(y) = H(F(y))$$

$$H(x) = x^2$$

$$F(y) = y.$$

$$G'(y) = 2(y) \cdot y'$$

$$2x + 2yy' = 0$$

Step 3. solve for y'

$$2x + 2yy' = 0$$

$$2yy' = -2x$$

$$y' = \frac{-2x}{2y} = -\frac{x}{y}.$$

$$P(3, 4)$$

 x_1, y_1

my slope.

$$m = y' \Big|_{(3, 4)} = -\frac{3}{4}.$$

Using slope-point.

$$y - y_1 = m(x - x_1)$$

$$y - 4 = -\frac{3}{4}(x - 3)$$

$$y = -\frac{3}{4}x + \frac{25}{4}.$$